

Aerodynamic Optimisation of Hovering Helicopter Rotors using Efficient and Flexible Shape Parameterisation

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A novel, generic and efficient shape parameterisation method is applied to CFD-based shape optimisation of helicopter rotors in hovering flight. Radial basis functions are used to provide an interpolation that transfers domain element movements into deformations of the design surface. The interpolation also extends to the CFD volume mesh, which is correspondingly deformed in a high-quality fashion. The shape parameterisation method requires very few design variables to allow free-form design. The method of shape and CFD mesh parameterisation is independent of mesh type (structured or unstructured), and optimisation independence from the flow solver (inviscid, viscous, aeroelastic) is achieved by obtaining sensitivity information for an advanced gradient-based algorithm by finite-difference. This has resulted in a flexible and versatile method of 'wrap-around' optimisation framework, which has also been parallelised.

The shape parameterisation and optimisation method has been applied previously to two-dimensional highly constrained aerofoil optimisations, which showed that significant drag reductions can be achieved. Recent three-dimensional fixed-wing optimisations have led to novel planform designs also with substantial performance benefits. In the present work, the method is extended to rotor blades, which will be subsequently optimised for improved performance (reduced torque) in hovering flight with rigid constraints in thrust, internal volume and root moments. The blades can be parameterised at various levels, focusing on gross planform deformations, local detailed surface changes, or combinations of both. These optimisations are currently underway for the ONERA 7A four-bladed case and the Caradonna and Tung two-bladed case.

I. Introduction

Optimisation is currently playing a key role in the aerospace industry, improving aerodynamic and structural designs within pre-specified limits and constraints. However, the quality and applicability of results gained by numerical optimisation techniques are inherently dependent on the fidelity of the analysis tool used. Computational fluid dynamics (CFD) is at the forefront of aerodynamic analysis capabilities, and application of numerical optimisation algorithms with such analysis has already produced notable results,^{1,2} the objective of which is to improve the numeric value of an objective function, defined as an indicator of aerodynamic performance, typically based on drag, lift and moment coefficients. Aerodynamic optimisation has been applied to rotors in hover,³ however, the analysis tools have often been of a low fidelity and only gross geometric changes can be analysed. Although CFD has been used in optimisations of two-dimensional aerofoil sections⁴⁻⁷ and even aspects of three-dimensional aircraft,⁸⁻¹² (and 3D aeroelastic aircraft^{13,14}) little has been achieved in the way of rotors. This is primarily due to high computational cost. High density and high quality grids are necessary to capture the vortical wake, and long solution integration times are required to develop and capture this wake when compared its fixed-wing counterpart.¹⁵⁻¹⁷ Hence, rotor blades place great strain on the computational power required to optimise such a system. Furthermore, there are two

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distinct flow regimes, hover and forward flight, and good performance in one is unlikely to be carried over to the other.

Although a CFD RANS optimisation of a rotor is presented by research at ONERA,^{18,19} it is achieved with extremely low density grids and incorporates a very small number of design variables, and so can be considered proof-of-concept research. The most advanced optimisation of the hovering rotor to date has been presented by Nadarajah,²⁰ where an adjoint-based optimisation of the Caradonna and Tung rotor has been accomplished. Although significant performance benefits have been achieved, only detailed aerofoil design can be altered, with planform remaining constant, and importantly, pitching moment has not been constrained.

Within the UK Rotorcraft DARP (Defence and Aerospace Research Partnership) a collaborative numerical simulation research theme exists between the Universities of Bristol and Liverpool and Agusta-Westland Helicopters. Liverpool is responsible for the flow-solver, an unsteady implicit, Navier-Stokes, parallel multi-block code,²¹ and Bristol for the mesh generation issues.²² The objective of this consortium is to use CFD in the main industrial design loop at Agusta-Westlands, and one aspect of this is the development of aerodynamic optimisation tools, to be wrapped around the flow-solver and grid generation software already under development. Hence, it is the objective of this paper to develop and present generic tools which allow optimisation of rotor blades in hover to be considered, these tools being independent of flow-solver, grid generator, and grid type. A domain element shape parameterisation method has been developed and presented recently for CFD-based shape optimisation in two dimensions,^{23,24} and is to be presented shortly for three-dimensional fixed-wing optimisations.²⁵ The objective of the research presented here is the extension to three-dimensional helicopter rotor blades of this versatile method for aerodynamic optimisation. The method achieves three aims:

- [1] Provide a generic 'wrap around' optimisation tool that is independent of both CFD flow solver and grid generation package.
- [2] Provide a method that allows high-fidelity aerodynamic optimisation of two- and three-dimensional bodies with a low number of design variables.
- [3] The parameterisation is extremely flexible and allows for a hierarchy of design variables; these can range from gross three-dimensional planform changes to fine detailed aerofoil section changes. This is a novel and unique aspect of the parameterisation method developed, such that detailed surface changes and large scale planform changes can be made to the three-dimensional design without the requirement for regeneration of a CFD volume mesh, and the resulting deformed volume mesh remains of high quality.^{26,27}

The parameterisation technique, surface mesh motion and volume mesh motion are all accomplished through the use of an interpolation using radial basis functions.

The shape parameterisation method is coupled to an advanced gradient-based constrained optimisation algorithm. This optimisation suite has recently been parallelised to allow optimisation of three-dimensional bodies in practical times. The sensitivity evaluation module has been parallelised to allow parallel evaluation of the required gradients, and this remains independent of the flow solver, so either a serial or parallel version of the flow solver itself may be called.

II. Domain Element Parameterisation

The novel parameterisation method links a set of aerodynamic mesh points to a domain element that controls the shape of the design. The domain element is automatically positioned around the exterior of the design. At the centre of this parameterisation technique is a multivariate interpolation using radial basis functions, and this provides an inverse mapping between the domain element, the surface geometry and the locations of grid points in the volume mesh. The inverse mapping is only required once for the initial design as the values of the parametric coordinates of the grid points with respect to the domain element remain constant throughout the optimisation. Updates to the geometry and the corresponding mesh are provided simultaneously by application of multivariate interpolation; this is extremely fast and efficient and results in very high quality mesh deformation.^{26,27}

Although radial basis functions have been used for mesh deformation techniques in shape optimisation²⁸ and structural-aerodynamic coupling,^{26,27,29} their use in the current research is not too strictly or uniquely

parameterise an aerodynamic shape, but to allow that initial shape and its corresponding aerodynamic mesh to deform in a relevant way to enable non-code and mesh-type specific aerodynamic shape optimisation. The interpolation method developed here requires no connectivity information, and can therefore be applied equally well to either structured or unstructured grid topologies. Domain element points and volume mesh points are simply treated as independent point clouds with the dependence matrix computed only once.

The general theory of RBFs is presented by Buhmann³⁰ and Wendland,³¹ and the method used here is comprehensively detailed in Allen and Rendall.^{26,27} It is sufficient here to state that the positions of the aerodynamic mesh points, given by the vectors $\mathbf{X}_a$, $\mathbf{Y}_a$ and $\mathbf{Z}_a$, can be directly computed by:

$$\mathbf{X}_a = \mathbf{A} \mathbf{a}_x = \mathbf{A} \mathbf{C}^{-1} \mathbf{X}_{DE} = \mathbf{H} \mathbf{X}_{DE} \quad (1)$$

$$\mathbf{Y}_a = \mathbf{A} \mathbf{a}_y = \mathbf{A} \mathbf{C}^{-1} \mathbf{Y}_{DE} = \mathbf{H} \mathbf{Y}_{DE} \quad (2)$$

$$\mathbf{Z}_a = \mathbf{A} \mathbf{a}_z = \mathbf{A} \mathbf{C}^{-1} \mathbf{Z}_{DE} = \mathbf{H} \mathbf{Z}_{DE} \quad (3)$$

where $\mathbf{X}_{DE}$, $\mathbf{Y}_{DE}$ and $\mathbf{Z}_{DE}$ are the vectors of domain element positions, $\mathbf{A}$ and $\mathbf{C}$ are matrices of basis functions, $\mathbf{a}_{x/y/z}$ are the vectors of coefficients, and $\mathbf{H}$ is the time-invariant dependence matrix.

Features of the RBF interpolation developed here worth noting are:

- The interpolation is independent of mesh type or structure
- The interpolation is unique.
- The interpolation in all three coordinate directions is independent.
- The interpolation is time-invariant, and so only needs to be computed once.
- The interpolation is perfectly parallel.

III. Optimisation Approach

When considering practical and relevant optimisation of aerodynamic performance of a solid body (aerofoil/ wing /rotor blade), there are usually constraints which need to be imposed (minimum thickness, minimum volume, minimum lift, maximum pitching moment etc). Unconstrained optimisations can incorporate constraints by using a penalty function for constraints that are near or beyond the constraint boundary. However, these methods are now considered inefficient and have been replaced by methods that focus on the solution of the Kuhn-Tucker equations. A Feasible Sequential Quadratic Programming (FSQP)³²⁻³⁴ algorithm is employed in the current research and represents the forefront of nonlinear programming techniques.

The aim of an optimisation process is to obtain the values of a set of design variables, $\underline{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n)^T$, that in some way can be defined as optimal. An objective function(s), determined by the numeric values of the design variables, $\underline{J}(\underline{\alpha})$, is to be minimised and can be subject to equality constraints, inequality constraints, and/or parameter bounds.

A general constrained problem description is stated as:

$$\text{minimise } \underline{J}(\underline{\alpha}) \quad (4)$$

Subject to:

$$\text{Equality constraints} \quad G_i(\underline{\alpha}) = 0, \quad i = 1, \dots, m_e$$

$$\text{Inequality constraints} \quad G_i(\underline{\alpha}) \leq 0, \quad i = m_e + 1, \dots, m$$

$$\text{Parameter bounds} \quad \underline{\alpha}_l \leq \underline{\alpha} \leq \underline{\alpha}_u$$

where the vector function $\underline{G}(\underline{\alpha})$ returns a vector of length m containing the values of the equality and inequality constraints evaluated at $\underline{\alpha}$.

Optimisation independence from the flow solver is achieved by obtaining sensitivity information by finite-differences using a symmetric central-difference stencil. The optimisation suite has recently been fully parallelised such that the evaluation gradient information can be obtained over multiple processors allowing optimisation in practical run times. The formulation and the parallel nature of the optimiser are presented in detail in Morris et al.³⁵

IV. Two-dimensional Rotor Aerofoil Optimisation

A rotor-type aerofoil case is considered. This is optimisation of an aerofoil section to provide a rotor with improved aerodynamic performance in forward flight. On the retreating side of a rotor blade at inboard locations it is often desirable to have an aerofoil that provides a large amount of lift at a significant incidence angle in low Mach numbers, but must also exhibit low drag at low incidence when on the forward side of the rotor disc. The NACA23012 aerofoil is a typical example of an aerofoil section used for this type of application. The optimisation of the NACA23012 aerofoil is presented with the objective to achieve increased lift at a high incidence at a low subsonic Mach number. The flow conditions $\text{Mach}=0.3$, $\text{Re}=3 \times 10^6$, $\alpha_I=12^\circ$ are used (α_I refers to the initial incidence of the aerofoil; note incidence can change through the optimisation as this is a design variable). A rigid constraint is placed on moment, volume, and importantly drag; this final constraint should ensure good working performance while the aerofoil is on the forward side of the rotor disc. Results of the optimisation are given in Table I, and initial and final domain elements and aerofoil geometries are shown in Figure 1.

	Initial	Optimised	%Diff
Cl	1.305	1.386	+5.98
Cm	0.0063	-0.0062	-
Volume	0.0797	0.0799	+2.5
Cd	0.0415	0.0415	+0.0

Table 1. NACA23012 aerofoil results

It is clear from the C_p distribution plots in Figure 2, that there is an increased region of low pressure when compared to the initial aerofoil, however, the suction peak has diminished somewhat. Therefore the optimised aerofoil has been able to improve upon the already well-suited NACA23012 by reducing viscous stresses associated with the strong suction peak allowing an increased lift by nearly 6% without any volume, moment or drag penalty.

The method has been extended to three-dimensional wing optimisation and results are presented in Morris et al.²⁵

V. Three-dimensional Aerodynamic Rotor Optimisation

Figure 3 shows the domain element parameterisation extended to the rotor. Given an initial rotor design, in this case an ONERA7A, a domain element is automatically placed around the blade surface conforming to the local chord and incidence. The domain element consists of 15 two-dimensional slices evenly distributed along the span. Design variables themselves are not the individual location of each domain element point (although they could be). Instead an intuitive set of applicable shape deformation design variables have been developed by inverse design techniques,²⁴ and each two-dimensional slice has 22 design variables associated with it - these control local surface changes. It is important to note that the parameterisation is fully three-dimensional; such that deformations due to a movement of the domain element two-dimensional slice smoothly extend in the spanwise direction as well, so that no linear interpolation is required between slices. Combined with this series of local surface changes are design variables that are truly three-dimensional by definition; these control planform changes including angle of attack, sweep distribution and anhedral/dihedral distribution. In total there exist a maximum of 388 design variables - considerably less than would be required by many other shape parameterisation techniques for such a geometry.

The advantages of the domain element method of parameterisation and obtaining sensitivity information via finite-difference are clear with the rotor optimisation problem. The flow solver HMB²¹ and a parallel structured multiblock upwind Euler code¹⁷ developed at Bristol University are under continual advancement; obtaining the gradient information via finite-difference allows for easy implementation of these updated flow solvers as they become viscous and/or aeroelastic (this also allows for the objective function to advance, incorporating both viscous and aeroelastic terms). Advances to grid generation techniques are also improving with the inclusion very soon of the helicopter fuselage, and again the generic nature of the optimisation techniques applied allow for easy incorporation of such improvements. The domain element method allows for multiple levels of design variable to be incorporated into different stages of the optimisation:

- [1] Global design variables; such as the coefficients of a polynomial twist or chord distribution.
- [2] Local coarse; local chord, twist, and thickness at sections where two-dimensional domain elements are located.
- [3] Local fine; fine surface design variables; for example leading edge control.

A feasibility study into the validity of evaluating sensitivity information via finite-difference for the hovering rotor system is presented for the most severe cases of design variables of type [2] and [3]. A point worth noting is the different levels of design variables that are achievable with the same parameterisation technique, and optimisation of a rotor with design variables of type [1] and [2] only will still encompass a broad range of the design space with very few design variables.

The high quality one million cell hover structured grid used for computation was generated by GHMB²² and shown in Figure 4a. The parameterisation method has been developed such that it is independent of both mesh type and importantly mesh size. This allows for the use of high quality and dense meshes that are typically required to obtain accurate hover flow solutions.

Sensitivity information for two design variables is presented; namely 1/4 chord twist (design variable type [2]) and thickness at 30% chord (design variable type [3]). An element section at approximately 2/3 tip radius was perturbed in each design variable separately by various amounts; grids remain of high quality, as shown in figure 4b and c. The steady-state inviscid hover solution provided by the Bristol parallel upwind structured multiblock code¹⁷ was obtained for each perturbation. A range of perturbation sizes is shown (Figure 7 and Figure 8) conforming to the different design variables due to their different nature (twist perturbations are presented in radians). Thrust coefficients vary linearly with perturbation size, Figure 7, and as expected design variable type [2] has a steeper gradient than design variable type [3], and the scaled sensitivity of thrust coefficient was found to be almost constant across the perturbation range for both design variables, Figure 8.

Figure 5 shows the vorticity plot for the ONERA 7A rotor and two plots for rotor blades perturbed at 2/3 tip radius in 1/4 chord twist (the perturbation is in a negative sense to increased angle of attack). Comparison shows that for a small perturbation, the flow field is only marginally altered, whereas for a larger perturbation a visible change in the vorticity in the wake sheet has occurred around the 2/3 tip radius position. Significantly both these perturbations give approximately the same sensitivity in thrust. These sentiments are repeated for λ_2 plots shown in Figures 6, where also observed for the higher perturbed rotor, the wake is not swept below the disk as fast; expected with the lower thrust coefficient it produces. This is an encouraging result demonstrating the feasibility for the finite-difference method in three-dimensional hovering rotor applications.

The optimisation of the hovering rotor is currently underway. The objective of which is to minimise the torque moment, subject to rigid constraints on thrust, pitching and bending aerodynamic moments and internal volume.

VI. Conclusions

A generic, unique and efficient domain element method of geometric parameterisation has been presented for application to CFD-based aerodynamic optimisation. The parameterisation uses radial basis functions to interpolate positions of the domain element and the grid coordinates to provide simultaneous deformation of the design surface and its corresponding CFD mesh.

A reduced number of design variables is key when obtaining sensitivity information via finite-difference, and having few parameterisation control points (in this case domain element nodes) is key to RBF interpolation for mesh deformation. A unique aspect of the parameterisation technique allows the use of design variables of different scales and types with only a few parameterisation nodes. These range from gross planform to detailed surface deformation degrees of freedom, and these can be combined to allow free-form design, but requiring a significantly reduced number of design variables for a three-dimensional application when compared to many other types of shape parameterisation method. The use of truly three-dimensional planform design variables is novel and has recently been shown to lead to innovative planform adjustments for fixed-wing cases in an effort to improve performance.

The methods have been applied previously to two-dimensional aerofoil optimisation and three-dimensional fixed-wing designs, achieving substantial performance improvements. However, the principal objective of

the current work is the extension to aerodynamic optimisation of hovering rotor blades, with the minimum number of design variables, while remaining independent of both the flow solver and grid generation package. The optimisation techniques presented make this possible. As a feasibility study, an ONERA 7A rotor blade with a one million cell mesh has been parameterised with the domain element method, and the inviscid sensitivity to thrust has been used to validate obtaining gradient information via finite-difference for such a system.

A parallel version of an advanced feasible sequential quadratic programming (FSQP) gradient-based optimiser has recently been developed, such that optimisation of rotors can be performed in a practical run times. The optimisation suite has been parallelised in a generic fashion such that the code is still independent of the flow solver. This enables a call to either a serial or parallel flow code for even faster computation.

Variable fidelity aerodynamic optimisation of the ONERA 7A four-bladed case, and the Caradonna and Tung two-bladed case, is currently underway, with minimum torque the objective, and strict constraints on thrust coefficient, root moments and internal volume.

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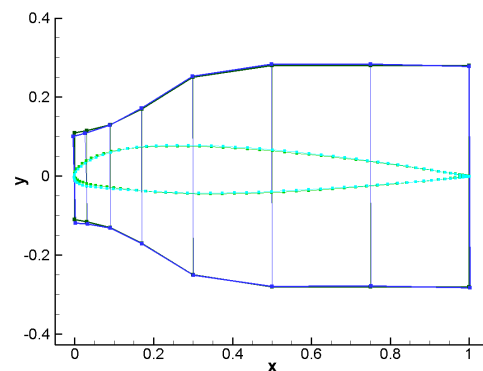


Figure 1. Domain element and aerofoil points: NACA23012 aerofoil -green and light green, optimised aerofoil -blue and light blue

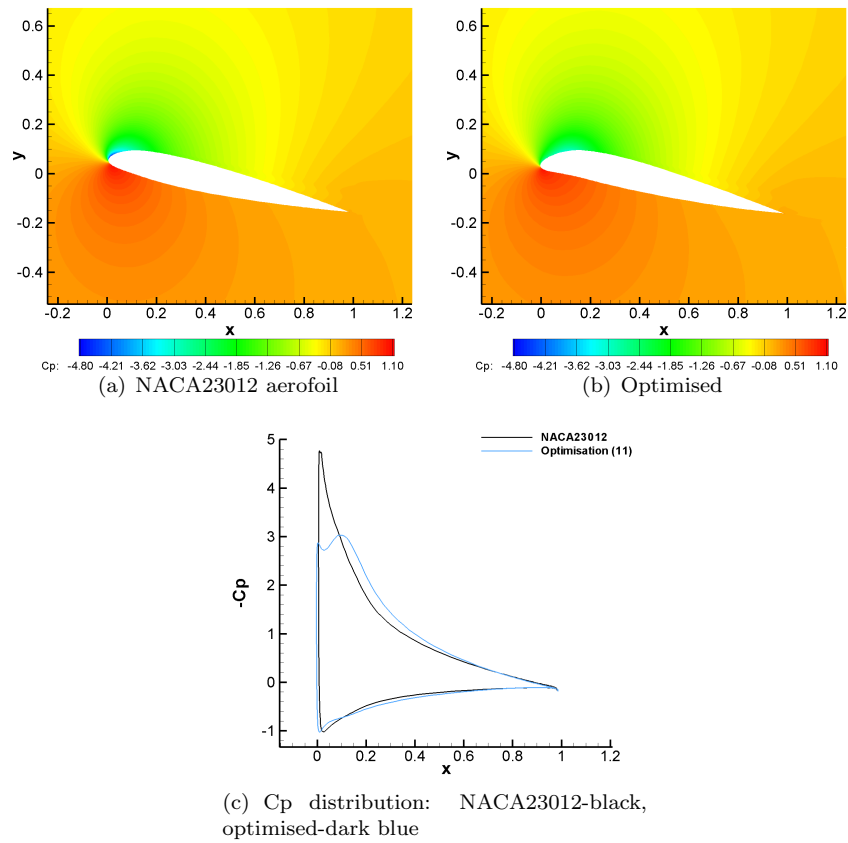
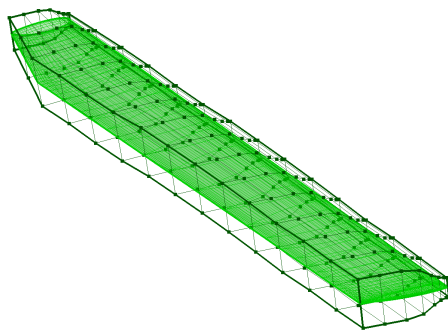
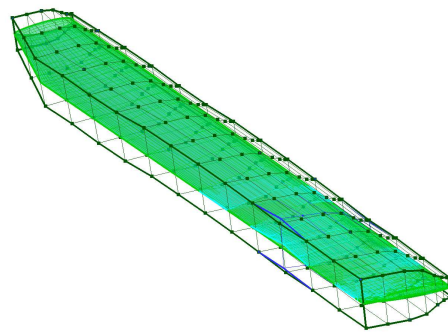


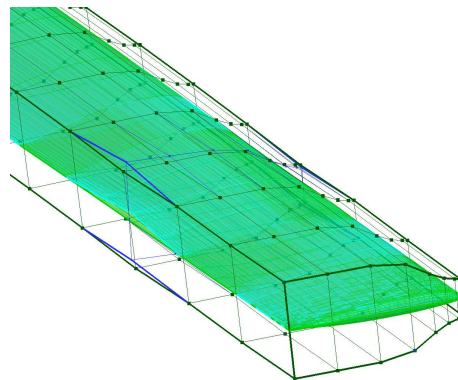
Figure 2. C_p plots of initial and optimised aerofoils



(a) ONERA7A rotor blade

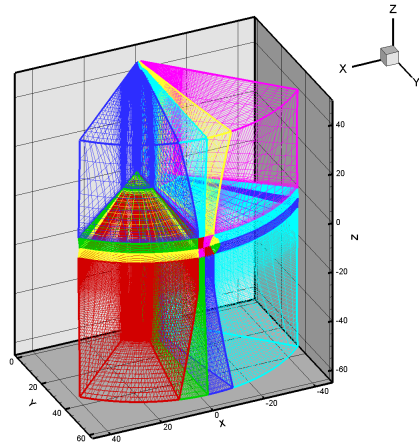


(b) ONERA7A rotor blade and 4^0 perturbation
in $1/4$ chord twist at $2/3$ radius

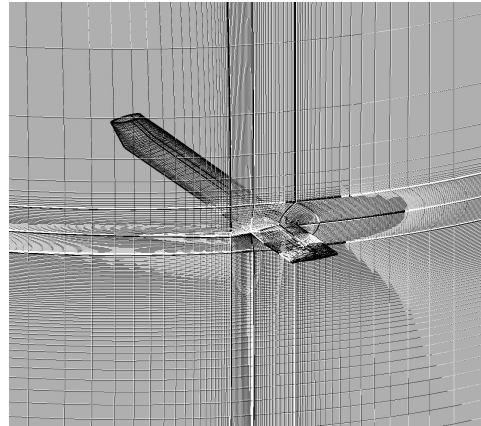


(c) ONERA7A rotor blade and 4^0 perturbation
in $1/4$ chord twist at $2/3$ radius

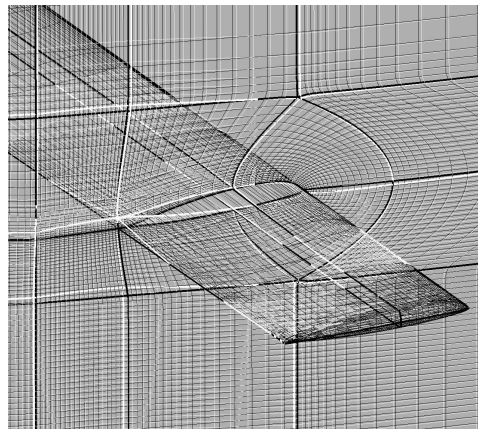
Figure 3. Domain element and blade surface mesh



(a) Structured 1.0 million point mesh of ONERA7A used for hover solutions



(b) Mesh deformation view (initial-black, 4th perturbation in 1/4 chord twist at 2/3 radius-white)



(c) Mesh deformation views (initial-black, 4th perturbation in 1/4 chord twist at 2/3 radius-white)

Figure 4. Domain element and blade volume mesh

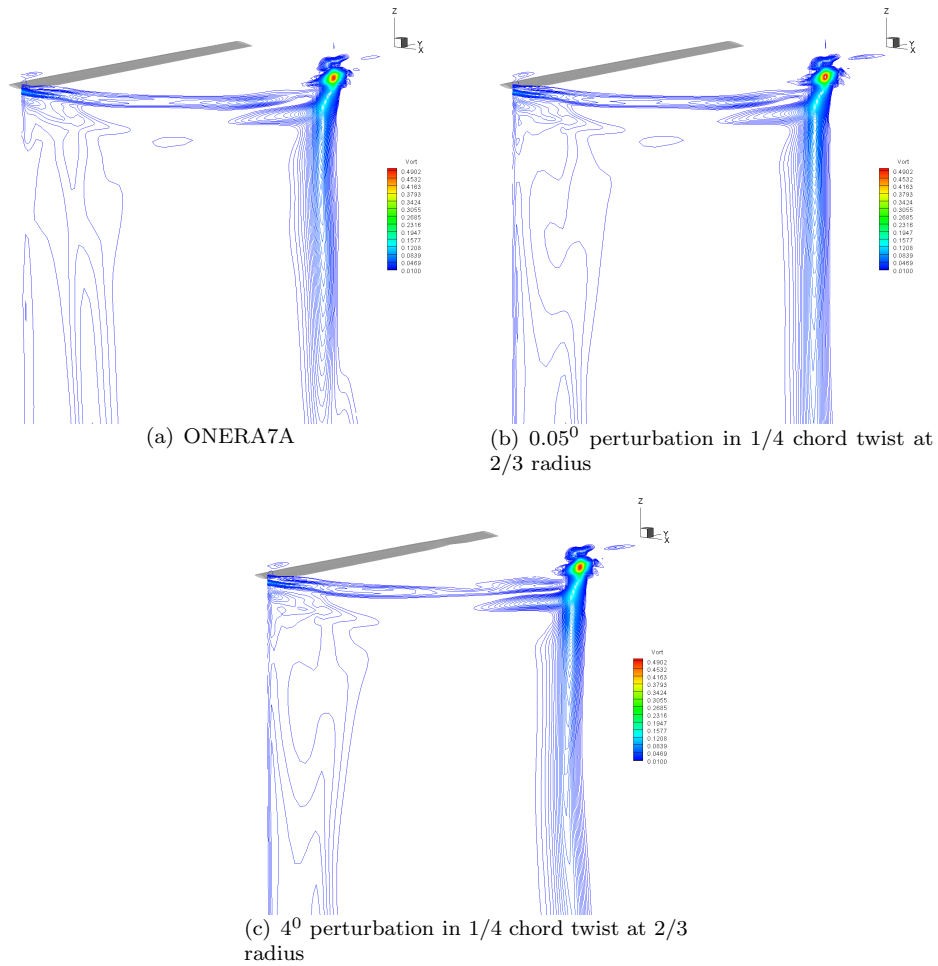
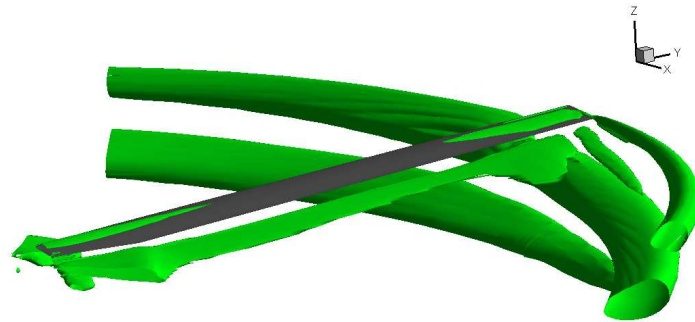
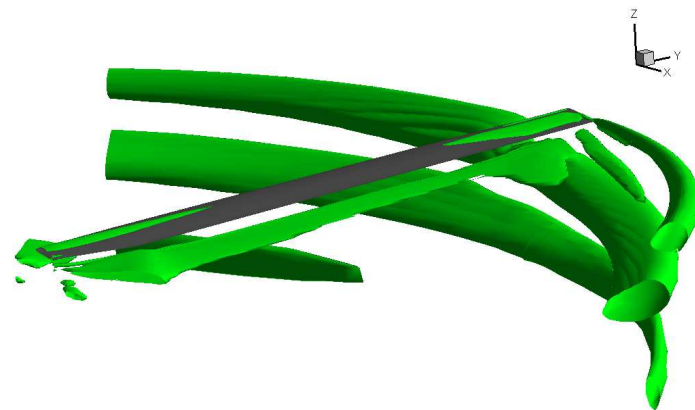


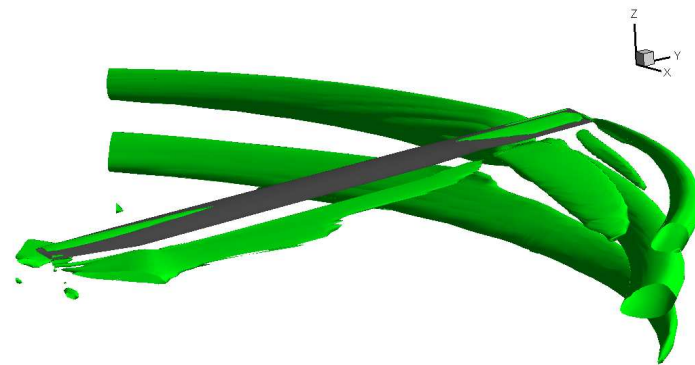
Figure 5. Vorticity at plane 45° behind blade



(a) ONERA7A



(b) 0.05° perturbation in 1/4 chord twist at 2/3 radius



(c) 4° perturbation in 1/4 chord twist at 2/3 radius

Figure 6. Lambda2 vorticity Plots (Lambda2=-0.0045)

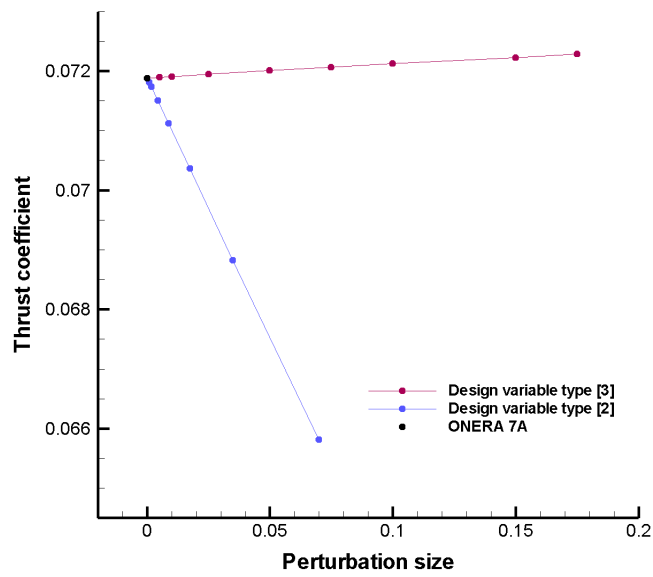


Figure 7. Variation in inviscid thrust coefficient of each design variable for differing perturbation sizing

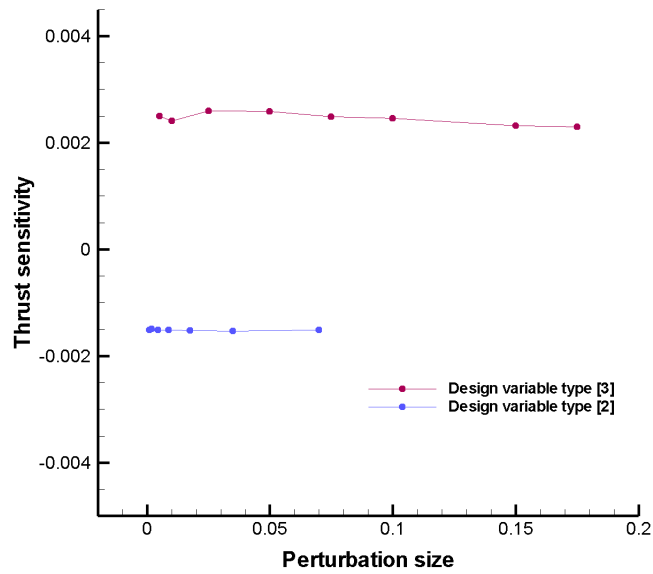


Figure 8. Inviscid thrust sensitivity of each design variable for differing perturbation sizing